

SOFTWARE ENGINEERING LAB

Lab 1: Requirements Engineering & UML Use-Case Modelling

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PROBLEM STATEMENT :#55

Book Club Reading Challenge & Discussion Portal

1. Introduction and scenario review

The Book Club Reading Challenge & Discussion Portal is a community reading platform designed to track members' personal reading page goals, facilitate chapter-specific discussions while protecting users from spoilers, and conduct monthly polls to select the next book.

Target Actors:

1. Club Member
2. Discussion Lead

Main Functions Identified:

1. Set reading goals
2. Track reading progress
3. Participate in chapter discussions
4. Manage/reveal spoiler-tagged content
5. Conduct monthly book polls
6. Vote for the next book

2. Elicit & Draft Requirements

Req ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall allow a Club Member to set and update a personal reading page goal for a selected book.	High	Pass: Member enters a valid page goal and the system saves and displays it. Fail: The goal is not saved or displayed.	Enables personalized reading challenges.
FR-002	Functional	The system shall allow a Club Member to record and view their reading progress against the page goal.	High	Pass: Member enters pages read and the system updates the progress correctly. Fail: The recorded progress remains unchanged.	Allows members to monitor their reading progress.
FR-003	Functional	The system shall allow Club Members to create and participate in chapter-specific discussion threads.	High	Pass: A Club Member can select a chapter and create or post a comment in its discussion thread. Fail: The comment cannot be associated with the selected chapter.	Provides the core community discussion functionality.
FR-004	Functional	The system shall hide spoiler-tagged discussion content until the Club Member explicitly chooses to reveal it or has marked the corresponding chapter as read.	High	Pass: Spoiler content remains hidden until the member chooses to reveal it or marks the chapter as read. Fail: Spoiler content is visible before either condition is satisfied.	Prevents members from accidentally seeing chapter spoilers.
FR-005	Functional	The system shall allow Club Members to cast one vote in each monthly poll for selecting the next book.	High	Pass: A Club Member can select one available book and submit one vote in the active monthly poll. Fail: The member cannot submit a valid vote or can submit multiple votes in the same poll.	Provides the mechanism for selecting the next book.
NFR-001	Non-Functional	The system shall prevent duplicate votes in a monthly poll by verifying the Club Member's session before recording a vote.	High	Pass: Repeated voting attempts from the same verified session are rejected and only one vote is recorded. Fail: More than one vote is recorded for the same member in the same monthly poll.	Maintains the integrity and fairness of monthly voting.
NFR-002	Non-Functional	The system shall display updated monthly poll vote tallies within 2 seconds after a valid vote is submitted.	Medium	Pass: The updated vote tally appears within 2 seconds after a valid vote is submitted. Fail: The updated tally takes more than 2 seconds to appear.	Provides timely feedback to members after voting.

4. Extract Actors & Use Cases:

ACTORS:

Actor 1 - Club Member

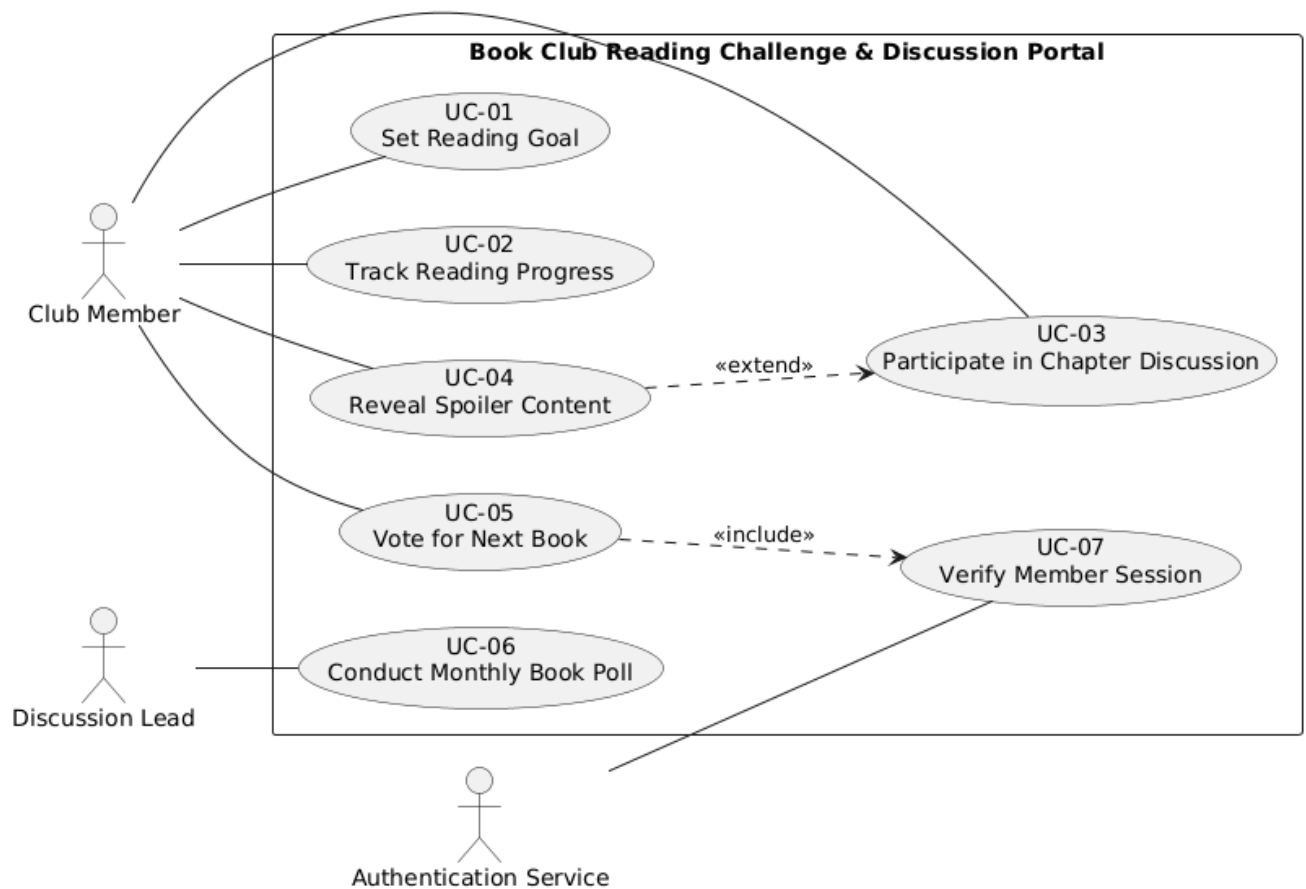
Actor 2 - Discussion Lead

Actor 3 – Authentication Service

USE CASES:

UC - 01	Set reading Goal	FR-001
UC - 02	Trach Reading progress	FR-002
UC - 03	Participate in Chapter Discussion	FR-003
UC - 04	Reveal Spoiler Content	FR-004
UC - 05	Vote for Next Book	FR-005
UC - 06	Verify Member Seesion	FR-005

5. UML Use-Case Diagram



6. Use-Case Flow Specification

Use Case ID: UC-05

Use Case Name: Vote for Next Book

Primary Actor: Club Member

Preconditions

1. The Club Member has an account in the system.
2. The Club Member is logged into the portal.
3. A monthly book poll is currently active.

Postconditions

1. The Club Member's vote is recorded successfully.
2. The poll tally is updated.
3. The Club Member receives a confirmation message.

Main Success Scenario

1. Club Member opens the active monthly book poll.
2. System displays the available book options.
3. Club Member selects one book.
4. Club Member submits the vote.
5. System verifies the Club Member's session.
6. System checks whether the Club Member has already voted in the current monthly poll.
7. System records the selected book as the Club Member's vote.
8. System updates the monthly poll tally.
9. System displays a confirmation message to the Club Member.
10. Use case ends successfully.

Alternate Flow – Duplicate Vote

6a. Club Member has already voted in the current monthly poll.

1. System detects that the Club Member has already submitted a vote.
2. System rejects the new vote.
3. System displays a message stating that only one vote is allowed per monthly poll.
4. The previously recorded vote remains unchanged.
5. Use case ends without recording another vote.